

SQL SERVER 2016

CHAPTER 06: HOW TO CODE SUBQUERIES

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Q1. _____ is used inside another query.

- A. An aggregate operator
- B. A join
- C. A subquery
- D. A scalar

Answer: C

Q2. Which one is **not** a valid option for using a subquery?

- A. In a WHERE clause as a search condition
- B. In the FROM clause as a table specification
- C. In the SELECT clause as a column specification
- D. In the ORDER BY clause as a sort order specification

Answer: D

[Ref. Page: 184]

Q3. When a subquery can be used anywhere in a SELECT statement where an expression is allowed?

- A. When the subquery returns a single value
- B. When a subquery returns a single column result set with one or more rows
- C. When a subquery returns a result set with two or more columns
- D. Above all cases

Answer: A

[Ref. Page: 184]

Q4. When a subquery can be used in the FROM clause in a SELECT statement?

- A. When the subquery returns a single value
- B. When a subquery returns a single column result set with one or more rows
- C. When a subquery returns a result set with two or more columns
- D. Above all cases

Answer: C

[Ref. Page: 184]

Q5. When a subquery can be used in the FROM clause in a SELECT statement in place of a list of values such as the list for an IN operator?

- A. When the subquery returns a single value
- B. When a subquery returns a single column result set with one or more rows
- C. When a subquery returns a result set with two or more columns
- D. Above all cases

Answer: B

[Ref. Page: 184]

Q6. Which one a subquery cannot return?

- A. a single value
- B. a result set with one or more rows
- C. a result set with two or more columns
- D. multiple result sets

Answer: D

[Ref. Page: 184]

Q7. Which one a subquery **cannot** include?

- A. WHERE clause
- B. GROUP BY and HAVING clause
- C. ORDER BY clause without TOP clause
- D. ORDER BY clause with TOP clause

Answer: C

[Ref. Page: 185]

Q8. Subqueries cannot be nested within another subqueries?

- A. True
- B. False

Answer: B

[Ref. Page: 185]

Q9. The outer query can include columns from the tables named in the subquery.

- A. True
- B. False

Answer: B

[Ref. Page: 187]

Q10. A query with join typically performs faster than the same query with a subquery.

- A. True
- B. False

Answer: A

[Ref. Page: 187]

Q11. A query that use NOT IN operator with a subquery can alternatively be restated using

-
- A. an inner join
 - B. an outer join
 - C. a cross join
 - D. a self-join

Answer: B

[Ref. Page: 189]

Q12. When can you use a comparison operator in a search condition to compare an expression with the result of a subquery?

- A. When the subquery returns a single result
- B. When the subquery returns a result set with one or more rows
- C. When the subquery returns a result set with two or more columns
- D. In all of the above case

Answer: A

[Ref. Page: 191]

Q13. Which one must be true when you use a subquery in WHERE clause with "=" operator?

- A. The subquery must return a single value
- B. The subquery must return a list of values
- C. The subquery must be correlated with outer query
- D. You cannot use a subquery in WHERE clause with "=" operator

Answer: A

Q14. Which operator cannot be used when a subquery in WHERE clause returns a list of values?

- A. IN
- B. EXISTS
- C. ANY

D. >

Answer: D

Q15. When can you use a comparison operator in a search condition to compare an expression with the result of a subquery that returns a list of values?

- A. When you include ANY or ALL or SOME keyword
- B. When you use DISTINCT keyword
- C. When you use TOP clause in the subquery
- D. None of the above

Answer: A

[Example

SELECT * Customers

WHERE Country = ANY

(SELECT DISTINCT Country FROM Employees)

Ref. Page: 191]

Q16. $X > ALL(1, 2)$ is equivalent to the expression _____.

- A. $X > 1$
- B. $X > 2$
- C. $X > 3$
- D. $X = 1.5$

Answer: B

[Ref. Page: 193]

Q17. $X = ALL(1, 2)$ is equivalent to the expression _____.

- A. $X = 1$
- B. $X = 2$
- C. $X = 1 \text{ AND } X = 2$
- D. $X = 1 \text{ OR } X = 2$

Answer: C

[Ref. Page: 193]

Q18. If no rows are returned by the subquery, a comparison that uses the ALL keyword is _____.

- A. True
- B. False
- C. NULL
- D. Unknown

Answer: A

[Ref. Page: 193]

Q19. If all rows returned by the subquery contains NULL, a comparison that uses the ALL keyword is _____.

- A. True
- B. False
- C. NULL
- D. 1

Answer: B

[Ref. Page: 193]

Q20. $X = ANY(1, 2)$ is equivalent to the expression _____.

- A. $X = 1$
- B. $X = 2$
- C. $X = 1 \text{ AND } X = 2$
- D. $X = 1 \text{ OR } X = 2$

Answer: D

[Ref. Page: 195]

Q21. Which one is equivalent to the SOME keyword?

- A. ALL
- B. ANY
- C. EXCEPT
- D. INTERSECT

Answer: B

[Ref. Page: 195]

Q22. If no rows are returned by the subquery or all of the rows returned by the subquery contains NULL, a comparison that uses the ANY keyword is _____.

- A. True
- B. False
- C. NULL
- D. Unknown

Answer: B

[Ref. Page: 195]

Q23. Consider the following two statements

Statement I: a correlated subquery is evaluated only once irrespective of how many rows the outer query returns.

Statement II: a correlated subquery is evaluated once for each row processed by the outer query.

Now choose the correct one about the above statements?

- A. Statement I is true but Statement II is false
- B. Statement I is false but Statement II is true
- C. Both Statement I and Statement II are false
- D. Both Statement I and Statement II are true

Answer: B

Q24. Which one you can use to check for the existence of an object before running a create statement?

- A. EXISTS
- B. IN
- C. ALL
- D. ANY

Answer: A

Q25. _____ is made up of the columns and rows of a result set from a query.

- A. A derived table
- B. A subquery
- C. Aggregate
- D. A correlated subquery

Answer: A

Q26. To create a derived table what criteria you must meet?

- A. enclose the query that generates the result set in parentheses
- B. alias the results of the query
- C. enclose the query that generates the result set in a UDF
- D. declare a variable table

Answer: A, B

Q26. A typical nested subquery is executed _____irrespective of how rows returned by the outer main query.

- A. Only once
- B. Once for each row processed by the outer query

-
- C. Once for each column processed by the outer query
 - D. Once for each time a NULL value is encountered by the outer query

Answer: A

[Ref. Page: 197]

Q27. A correlated subquery is executed _____.

- A. Only once
- B. Once for each row processed by the outer query
- C. Once for each column processed by the outer query
- D. Once for each time a NULL value is encountered by the outer query

Answer: B

[Ref. Page: 197]

Q28. Which operator is used to test whether or not a subquery returns a result set?

- A. ALL
- B. ANY
- C. EXIST
- D. IN

Answer: C

[Ref. Page: 198]

Q29. Which one is one referred to as derived table?

- A. A subquery that returns a single value and used as an expression in a SELECT statement
- B. A subquery that returns a single column result set with one or more rows and used in the search condition of a SELECT statement
- C. A subquery that returns a result set and used in the FROM clause of a SELECT statement
- D. A correlated subquery

Answer: C

[Ref. Page: 201]

Q30. Which one is common table expression?

- A. A query that returns a single value and used as an expression in a SELECT statement
- B. A query that returns a single column result set with one or more rows and used in the search condition of a SELECT statement
- C. A query that returns a result set and in the FROM clause of a SELECT statement
- D. An expression, usually a SELECT statement, that create a temporary table and is used in the following query

Answer: D